CHUNK_ID: hardware-layout
TITLE: Two modules: pressing and turning over
ORIGIN: REFERENCE
SOURCE: Anleitung - Festo CP Lab.pptx, slides 4-6; Festo CP & Module.pptx, slide 1

CONTENT
The CP Lab has two connected application modules: a press and a gripper that turns the workpiece over by 180 degrees. Turning means reorientation, not lathe machining. Conveyors, carrier stoppers, RFID and position sensors support transport. The press application uses a Revolution Pi; the turning-over module uses a microcontroller. Both communicate with a Siemens PLC. This reference is not a current connectivity test.
END CONTENT

CHUNK_ID: opc-evidence
TITLE: OPC interfaces and limits of observations
ORIGIN: REFERENCE
SOURCE: Anleitung - Festo CP Lab.pptx, slides 8-9; a4s/machines/festo_cp_lab/model.py

CONTENT
The model names Turning PLC, Press PLC and Press RevPi OPC servers. Application flags include xStart, xReady, xBusy and xError; station collective fault xSF is a separate signal. Previous project tests found xError could stay false during a station fault. RevPi Pressure_1/2 and Flow_1/2 are uncalibrated raw readings in this integration. Never compare these raw counts to the fictional bar thresholds in the demo. A connection timeout means unavailable evidence, not a physical stop.
END CONTENT

CHUNK_ID: readiness-reference
TITLE: Readiness depends on application communication
ORIGIN: REFERENCE
SOURCE: Anleitung - Festo CP Lab.pptx, slides 17-18

CONTENT
The commissioning reference mentions compressed-air supply and application communication readiness. At power-up the PLC may initially lack a connection to the RevPi or microcontroller. A qualified operator must use approved site procedures. These observations are possible checks, not proof of an air leak and not permission for an agent to reset, start or change physical equipment.
END CONTENT